

ddos_flood_evidence_test.sh — §11.4.18 companion doc

Revision: 1 Last modified: 2026-07-01T00:00:00Z

Overview

tests/regression/ddos_flood_evidence_test.sh is the §11.4.135 standing regression guard for finding **F5** — the DDoS / load-flood suite (tests/dynamic/suites/ddos_flood_suite.sh) used to score a "survived the flood / degraded-not-collapsed" **PASS without any positive captured evidence that a flood was ever generated**. It is a fixture-driven guard: it extracts the real pure classifier flood_survival_verdict() from the tracked suite and drives it with canonical fixtures. It needs **no live stack, no network, and no containers**.

Prerequisites

- POSIX sh, awk, mktemp, and bash (the extracted-function probe runner).
- The tracked suite tests/dynamic/suites/ddos_flood_suite.sh containing the flood_survival_verdict() function (the fix under guard).

Usage examples

```
# GREEN guard (default) – asserts the FIXED classifier refuses the
    zero-flood
# bluff, PASSES a real survived flood, still FAILs a real crash,
    and honestly
# SKIPs when the proxy is absent.
tests/regression/ddos_flood_evidence_test.sh

# RED reproduce – asserts the PRE-FIX survival gate (pid_stable=1
    && rec=200
# ONLY, ignoring the flood counter) PASSES a zero-flood run (the
    §11.4.1 bluff).
RED_MODE=1 tests/regression/ddos_flood_evidence_test.sh
```

Run under host-safety caps per the constitution:

```
GOMAXPROCS=2 nice -n 19 ionice -c 3 tests/regression/  
ddos_flood_evidence_test.sh
```

Exit code: 0 = PASS, 1 = FAIL. A structured verdict line plus an evidence file under qa-results/regression/ddos_flood_evidence/ are produced on every run.

The four GREEN fixtures

```
flood_survival_verdict <pid_stable> <recovery_code> <flood_total>  
<flood_responses> <proxy_listening>
```

Fixture	Args	Expected verdict	Why
ZERO	1 200 0 0 yes	FAIL:no-flood-evidence	Proxy up but the flood issued zero requests → the "survived a flood" claim is vacuous (§11.4.1 bluff refused).
SURVIVED	1 200 3000 2950 yes	PASS	A real flood (3000 issued, 2950 measurable), PID stable, recovered to 200.
CRASHED	0 000 3000 100 yes	FAIL:crashed-or-no-recovery	A real flood but PID changed / no recovery — the anti-bluff crash catch preserved.
ABSENT	1 200 0 0 no	SKIP:topology_unsupported	No flood evidence AND the proxy is not listening → honest topology SKIP, never a silent PASS.

Edge cases

- **Non-numeric counters** — flood_survival_verdict sanitises flood_total and flood_responses to 0 when empty/non-numeric

(case ... *`[!0-9]*`), so a malformed counter degrades to "no flood evidence" (FAIL/SKIP), never a bluff PASS.

- **Multi-line probe output** — the guard matches with case ... in `*A*B*` globs; * spans newlines in POSIX case, so the ordered four-fixture assertion holds.
- **RED cannot reproduce** — if the pre-fix replica does NOT PASS the zero-flood fixture, the guard reports a §11.4.7 finding rather than a false GREEN.

Internal behaviour

The guard writes a small probe script to a mktmp file:

- **GREEN** — awk-extracts the `flood_survival_verdict()` body verbatim from the tracked suite (real function, not a grep of expected strings), appends four `echo "<LABEL>=$ (flood_survival_verdict ...)"` drivers, and runs it under bash. It PASSES iff the concatenated output matches `*ZERO=FAIL:no-flood-evidence*SURVIVED=PASS*CRASHED=FAIL:crashed-or-no-recovery*ABSENT=SKIP:topology_unsupported*`.
- **RED** — emits the faithful pre-fix replica `_prefix_flood_verdict() { [ "$1" = 1 ] && [ "$2" = 200 ] && echo PASS || echo FAIL; }` and asserts it prints `ZERO=PASS` for the zero-flood fixture (the reproduced bluff).

§1.1 paired mutation

Mutating the suite's `flood_survival_verdict` so the zero-count case wrongly PASSES (e.g. neutralising the `flood_total<=0 || flood_responses<=0` guard) makes the GREEN run's `ZERO=FAIL:no-flood-evidence` expectation fail — the guard emits `REGRESSION: ...` and exits 1 with a real assertion mismatch (not a parse error). Restoring the suite byte-identical (md5 match) returns the guard to GREEN.

Related scripts

- `tests/dynamic/suites/ddos_flood_suite.sh` — the suite under guard (the fix).
- `tests/lib/evidence.sh` — `ab_pass_with_evidence / ab_skip_with_reason / port_is_listening` helpers the suite reuses.
- `tests/regression/external_egress_verdict_test.sh` — the house pattern this guard is modelled on.

Last verified: 2026-07-01